

The Problem of Abundance

How is having a lot of a necessary substance (here, food) a problem? The problem rarely is the surplus itself. If, as a farmer, I grow way more onions than anyone could ever eat, I can just leave them unharvested and plow them back into the soil; they will decompose over time. The “problem” with abundance is a social one: How do I benefit from a surplus that I cannot use myself? And how can my extended family, friends, or community benefit from that surplus in ways that ultimately benefit me, as well?

As described by Gremillion, biologists have studied how animals deal with abundant food sources, whether an overall good year for a given resource (e.g., for fruit or acorns) or whether a large food package is procured that is larger than the individual can consume (e.g., a chimpanzee catching a monkey). Animals may share in exchange for a variety of benefits, including better access to mates, that accrue directly to the individual doing the sharing. They may also share when the cost of defending a resource is more than the gain from keeping it—a phenomenon termed tolerated theft. In some cases, animals can store food for later. Humans, however, have developed social mechanisms for using surplus that build on these simple strategies, resulting in complex technological and social solutions.

Possible Solutions to Abundance

One simple solution to abundance is storage. If you have more food than you need today, but you may not have enough tomorrow, and it is possible to keep the extra from today for tomorrow, then storing your extra food is a good idea. This makes sense if you have something dry and durable, such as a handful of nuts, but less sense if you have highly perishable food, such as fish, and no refrigeration. Fresh fruit only remains edible for so long after it has been picked; refrigeration helps but can't keep produce in perfect shape forever. Thus, strategies for storage often include ways to transform a perishable food into a less perishable form. One simple example is dehydration: Dried fruit or meat jerky can last weeks or months at room temperature. More complex forms of food storage involve microbes as agents of transformation, producing fermented foods—such as bread, cheese, pickles, and alcohol—that are ubiquitous today. We will get much deeper into fermentation later in this module!

An alternative solution to abundance is to share food when there is a surplus. This food is converted into another currency: the recipient's obligation to the giver. Imagine a friend who would always come to your house with a bottle of wine, a loaf of freshly baked bread, or a bouquet of flowers for you. Would you feel like you were expected to return the favor with a gift when you visited them? Disproportionate giving and receiving creates asymmetrical relationships over time,

with an obligation to reciprocate with something greater than a simple gift. In such situations, differences in status and power can arise, as we will discuss more in the context of feasting later in this module.

Before we explore these solutions to abundance in more detail, however, let's understand more clearly what archaeologists look for when they try to identify abundance in the past. How do archaeologists identify a food as surplus, and how do they quantify the scale of that surplus?